

The Extended Euclidean Algorithm: Finding Modular Inverses for RSA

Why RSA needs it. To build an RSA key, we choose e and then must find d with

$$e \cdot d \equiv 1 \pmod{\varphi(n)}.$$

That d is the *modular inverse* of e . For real keys $\varphi(n)$ has hundreds of digits, so we cannot guess d or try all values. The Extended Euclidean Algorithm (EEA) computes d in a few thousand simple steps, each just an integer division. This note builds the algorithm from scratch in four stages:

1. the ordinary Euclidean algorithm (computes the gcd);
2. Bézout's identity (the gcd is a combination of the inputs);
3. the extended algorithm (tracks that combination in a table);
4. using it to get the inverse, with a full RSA example.

1. Warm-up: the ordinary Euclidean algorithm

Picture it

Suppose you have a 240×46 rectangle of floor and want to tile it with the largest possible square tiles, with no gaps and no cutting. Lay the biggest squares you can (46 by 46), and a strip is left over; now tile that leftover strip with the biggest squares that fit, and so on. The side of the last square that leaves nothing over is the gcd. Euclid's algorithm is exactly this “keep tiling the leftover” process.

Lemma 1. For integers $a \geq b > 0$, write $a = qb + r$ with $0 \leq r < b$ (division with remainder). Then $\gcd(a, b) = \gcd(b, r)$.

What it says, and why we need it. Divide a by b and keep the remainder r . The lemma says the new, smaller pair (b, r) has the same gcd as the old pair (a, b) . This is our licence to replace (a, b) by (b, r) in the algorithm; without it, Euclid's algorithm would be invalid.

Idea of the proof. Show that the two pairs have exactly the *same set of common divisors*. Two equal sets have the same largest element, and the largest common divisor is the gcd. To show two sets are equal we prove each is contained in the other, i.e. two directions. Both use only the equation

$$a = qb + r \quad \Longleftrightarrow \quad r = a - qb.$$

Proof. Direction 1: every common divisor of (a, b) is a common divisor of (b, r) . Let $d \mid a$ and $d \mid b$. Since $d \mid b$, also $d \mid qb$ (a divisor of b divides every multiple of b). So d divides both a and qb , hence divides their difference $a - qb = r$. Together with $d \mid b$, d is a common divisor of b and r .

Direction 2: every common divisor of (b, r) is a common divisor of (a, b) . Let $d \mid b$ and $d \mid r$. Then $d \mid qb$, so d divides the sum $qb + r = a$. Together with $d \mid b$, d is a common divisor of a and b .

Conclusion. The two directions give

$$\{\text{common divisors of } a, b\} = \{\text{common divisors of } b, r\}.$$

Equal sets have equal greatest elements, so $\gcd(a, b) = \gcd(b, r)$. $\square$

Why it is “obvious” in hindsight. Since a and r differ by a multiple of b , any number that divides b cannot tell a and r apart: it divides both or neither. So, among divisors of b , the ones that divide a are exactly the ones that divide r .

Concrete check: $a = 240$, $b = 46$. Here $240 = 5 \cdot 46 + 10$, so $q = 5$, $r = 10$.

Pair	Divisors of each number	Common divisors
(240, 46)	240: 1, 2, 3, 4, 5, 6, 8, 10, 12, 15, 16, 20, 24, 30, 40, 48, 60, 80, 120, 240; 46: 1, 2, 23, 46	1, 2
(46, 10)	46: 1, 2, 23, 46; 10: 1, 2, 5, 10	1, 2

The sets agree, so $\gcd(240, 46) = \gcd(46, 10) = 2$.

Small questions students often ask.

- *Why does $d \mid b$ imply $d \mid qb$?* If $b = dk$, then $qb = d(qk)$.
- *Why do $d \mid a$ and $d \mid qb$ imply $d \mid (a - qb)$?* If $a = dm$ and $qb = dn$, then $a - qb = d(m - n)$. Sums work the same way.
- *Why do equal sets give equal gcds?* The gcd is by definition the greatest element of the set of common divisors.
- *Is $r < b$ needed here?* Not for the lemma itself: it holds for any $r = a - qb$. The condition $0 \leq r < b$ is needed only so that the remainders keep shrinking, which is what makes the algorithm stop (next paragraph).

So we may replace (a, b) by the smaller pair (b, r) without changing the gcd, and repeat. The remainders strictly decrease, so we reach remainder 0; the last nonzero remainder is the gcd, because $\gcd(g, 0) = g$.

Example: $\gcd(240, 46)$.

$$240 = 5 \cdot 46 + 10$$

$$46 = 4 \cdot 10 + 6$$

$$10 = 1 \cdot 6 + 4$$

$$6 = 1 \cdot 4 + 2$$

$$4 = 2 \cdot 2 + 0$$

The last nonzero remainder is 2, so $\gcd(240, 46) = 2$.

How fast? The number of division steps grows only in proportion to the number of digits of the inputs (Lamé’s theorem). For 2048-bit numbers that is at most a few thousand steps — instant for a computer.

2. Bézout's identity: the gcd is a combination

Theorem 1 (Bézout). *For integers a, b (not both zero) there exist integers s, t with*

$$s a + t b = \gcd(a, b).$$

Look back at the example: the first line says $10 = 240 - 5 \cdot 46$. That is 10 written as a combination of 240 and 46. The second line gives $6 = 46 - 4 \cdot 10$, and substituting the previous line turns this into another combination of 240 and 46. Every remainder we produce can be rewritten as a combination of the two original numbers, and the last nonzero remainder is the gcd. That is the whole proof, and it is also the algorithm: *keep track of the combination for each remainder as you go*.

Back-substitution (by hand). Work upward from the last nonzero remainder in the example above:

$$\begin{aligned} 2 &= 6 - 1 \cdot 4 \\ &= 6 - 1 \cdot (10 - 1 \cdot 6) = 2 \cdot 6 - 1 \cdot 10 \\ &= 2 \cdot (46 - 4 \cdot 10) - 1 \cdot 10 = 2 \cdot 46 - 9 \cdot 10 \\ &= 2 \cdot 46 - 9 \cdot (240 - 5 \cdot 46) = -9 \cdot 240 + 47 \cdot 46. \end{aligned}$$

Check: $-2160 + 2162 = 2$. So $s = -9$, $t = 47$. This works, but it is messy and error-prone for long chains. The table method below does the same job going *forward*, with no backtracking.

3. The table method

The recipe

Start with the two numbers $a \geq b$ you are given. Build a table with columns i , q_i , r_i , s_i , t_i .
Two starting rows (memorize these):

$$r_0 = a, \quad s_0 = 1, \quad t_0 = 0 \qquad r_1 = b, \quad s_1 = 0, \quad t_1 = 1.$$

Every next row uses the two rows just above it:

$$q_i = \left\lfloor \frac{r_{i-2}}{r_{i-1}} \right\rfloor, \quad r_i = r_{i-2} - q_i r_{i-1}, \quad s_i = s_{i-2} - q_i s_{i-1}, \quad t_i = t_{i-2} - q_i t_{i-1}.$$

Stop when a remainder $r_i = 0$. The row just before it holds $\gcd(a, b) = r_{i-1} = s_{i-1} a + t_{i-1} b$.

Notice that r, s, t all follow the *same* update rule with the same q_i ; only the starting values differ. So each new row is: compute q from the two r 's, then do the same “previous-previous minus q times previous” in three columns.

The key invariant. In every row,

$$r_i = s_i a + t_i b$$

It holds for the starting rows ($a = 1 \cdot a + 0 \cdot b$ and $b = 0 \cdot a + 1 \cdot b$). If it holds for two consecutive rows, subtracting q_i times one equation from the other shows it holds for the next row. So it holds for every row, including the one with $r = \gcd$. *This is your built-in error check:* at any row you can verify $s_i a + t_i b = r_i$.

Example (same numbers as before): $a = 240$, $b = 46$.

i	q_i	r_i	s_i	t_i
0	—	240	1	0
1	—	46	0	1
2	5	10	1	-5
3	4	6	-4	21
4	1	4	5	-26
5	1	2	-9	47
6	2	0	23	-120

Row 5 is the last one before the remainder hits 0: $\gcd = 2$ and $2 = -9 \cdot 240 + 47 \cdot 46$, matching the back-substitution. Let us see one row being computed. Row 3 uses rows 1 and 2: $q_3 = \lfloor 46/10 \rfloor = 4$, then $r_3 = 46 - 4 \cdot 10 = 6$, $s_3 = 0 - 4 \cdot 1 = -4$, $t_3 = 1 - 4 \cdot (-5) = 21$. Check the invariant: $-4 \cdot 240 + 21 \cdot 46 = -960 + 966 = 6$. ✓

(The final row's s, t , here 23 and -120, equal $\pm b/g$ and $\mp a/g$; they are not used.)

4. From Bézout to the modular inverse

Theorem 2. Let $m > 1$ and $\gcd(x, m) = 1$. Run the EEA on $(a, b) = (m, x)$. If the row with $r = 1$ is $1 = sm + tx$, then

$$x^{-1} \equiv t \pmod{m}.$$

If $\gcd(x, m) \neq 1$, no inverse exists.

Proof. Reduce $1 = sm + tx$ modulo m : the term sm vanishes, leaving $1 \equiv tx \pmod{m}$. That is exactly the definition of t being the inverse of x . If $g = \gcd(x, m) > 1$, then g divides $tx + sm$ for every s, t , so $tx \equiv 1$ is impossible. $\square$

Practical shortcuts for inverses.

- Put the *modulus first*: $a = m$, $b = x$. Then t is the inverse and the s column is unnecessary — you may leave it out to save work.
- If t comes out negative, add m (repeat until it lies in $0, \dots, m-1$).

Example A (small, by hand): the inverse of 7 modulo 40

i	q_i	r_i	s_i	t_i
0	—	40	1	0
1	—	7	0	1
2	5	5	1	-5
3	1	2	-1	6
4	2	1	3	-17
5	2	0	-7	40

The highlighted row says $1 = 3 \cdot 40 + (-17) \cdot 7$. Hence $7^{-1} \equiv -17 \equiv 40 - 17 = \mathbf{23} \pmod{40}$.

Check: $7 \cdot 23 = 161 = 4 \cdot 40 + 1$. ✓

Example B (RSA): find d for $e = 17$ and $\varphi(n) = 3120$

Take the textbook key $p = 61$, $q = 53$: $n = 3233$ and $\varphi(n) = 60 \cdot 52 = 3120$. We need $d = 17^{-1} \bmod 3120$. First confirm $\gcd(17, 3120) = 1$; the table will do that for us. Run the EEA with $a = 3120$ and $b = 17$:

i	q_i	r_i	s_i	t_i
0	—	3120	1	0
1	—	17	0	1
2	183	9	1	-183
3	1	8	-1	184
4	1	1	2	-367
5	8	0	-17	3120

Reading the table, step by step.

- Row 2: $3120 = 183 \cdot 17 + 9$, so $q_2 = 183$, $r_2 = 9$; $s_2 = 1 - 183 \cdot 0 = 1$; $t_2 = 0 - 183 \cdot 1 = -183$.
- Row 3: $17 = 1 \cdot 9 + 8$, so $q_3 = 1$, $r_3 = 8$; $s_3 = 0 - 1 \cdot 1 = -1$; $t_3 = 1 - 1 \cdot (-183) = 184$.
- Row 4: $9 = 1 \cdot 8 + 1$, so $q_4 = 1$, $r_4 = 1$; $s_4 = 1 - 1 \cdot (-1) = 2$; $t_4 = -183 - 1 \cdot 184 = -367$.
- Row 5: $8 = 8 \cdot 1 + 0$. The remainder is 0, so stop; the row above (row 4) has $r = 1 = \gcd$.

Row 4 gives $1 = 2 \cdot 3120 + (-367) \cdot 17$. Check: $6240 - 6239 = 1$. ✓

So $17^{-1} \equiv -367 \pmod{3120}$, and adding 3120 gives

$$d = 3120 - 367 = \mathbf{2753}.$$

Check: $17 \cdot 2753 = 46801 = 15 \cdot 3120 + 1$. ✓

Using the key. Encrypt $m = 65$: $c = 65^{17} \bmod 3233 = 2790$. Decrypt: $2790^{2753} \bmod 3233 = 65$. The whole security of the private key rests on d being computable only if you know $\varphi(n)$, that is, only if you can factor n .

Example C: when there is no inverse

Try to invert 18 modulo 60:

i	q_i	r_i	s_i	t_i
0	—	60	1	0
1	—	18	0	1
2	3	6	1	-3
3	3	0	-3	10

The last nonzero remainder is $6 \neq 1$, so $\gcd(18, 60) = 6$ and 18 has no inverse mod 60. This is why RSA requires $\gcd(e, \varphi(n)) = 1$: if the table ends with a gcd bigger than 1, choose a different e .

Common pitfall

- **Order matters for reading t .** With $(a, b) = (m, x)$ the inverse of x is the t column. If you swap the order, the inverse of x is in the s column instead.
- **Stop at the right row.** The answer is in the row *before* the remainder 0, not in the row where $r = 0$.
- **Negative answers are fine.** $t = -367$ means $-367 \equiv 2753 \pmod{3120}$; add the modulus.
- **Always verify.** Multiply your inverse by x and reduce; you must get 1.

5. Practice problems

Use the table method for each. Answers are at the end.

1. Find $11^{-1} \pmod{26}$.
2. Find $13^{-1} \pmod{60}$.
3. Does 30 have an inverse modulo 84? Justify.
4. For the key $p = 11$, $q = 13$, $e = 7$: compute $\varphi(n)$ and then $d = e^{-1} \pmod{\varphi(n)}$.

Answers. (1) 19, since $11 \cdot 19 = 209 = 8 \cdot 26 + 1$. (2) 37, since $13 \cdot 37 = 481 = 8 \cdot 60 + 1$. (3) No: $\gcd(30, 84) = 6 \neq 1$. (4) $\varphi(143) = 10 \cdot 12 = 120$, and $d = 103$, since $7 \cdot 103 = 721 = 6 \cdot 120 + 1$.

Where else this appears in the course

The same computation is used, with both s and t , in the *Common Modulus Attack*: with $\gcd(e_1, e_2) = 1$ the EEA gives a, b with $ae_1 + be_2 = 1$, letting an attacker combine two ciphertexts into the plaintext. In the *Shared-Prime Attack* the plain Euclidean algorithm (Section 1 of this note) finds the common prime factor of two moduli. In RSA key generation, the EEA is the step that turns the public exponent e into the private exponent d .